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Tobic : Wireshark

What is wireshark?

Wireshark is a free, open-source network protocol analyzer that captures the data packets travelling across a network and shows them in a readable form. It displays who sent each packet, who received it, which protocol it used (such as DNS, HTTP, TCP, or TLS), and what data it carried.

Step 1: Install Wireshark

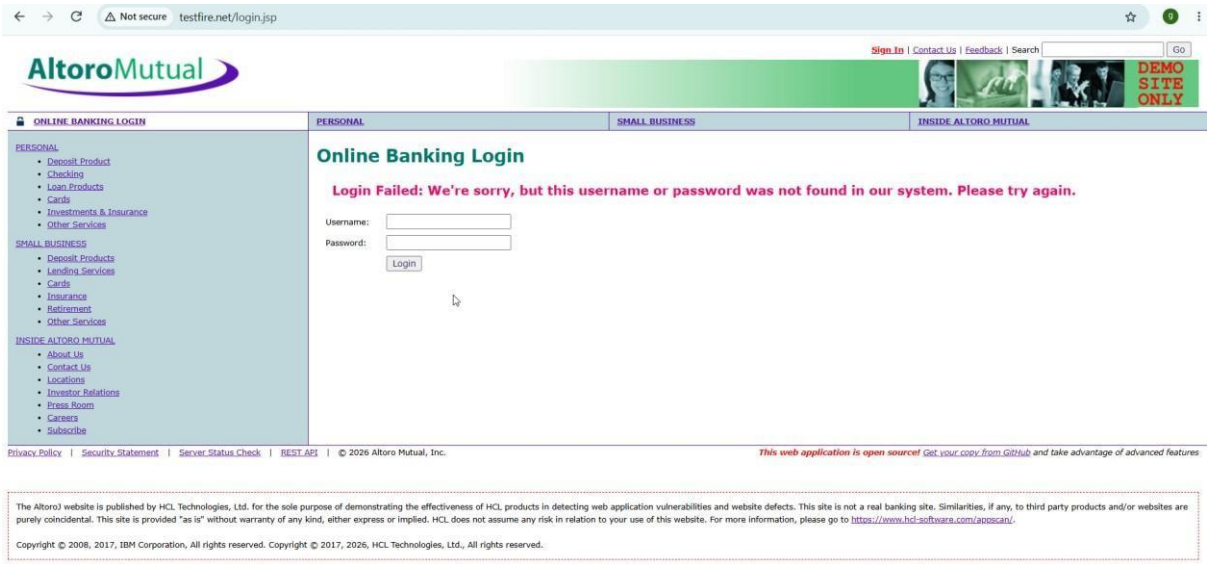
1. Download Wireshark from the official website.
2. Run the installer and accept the defaults, including Npcap (Windows).
3. Complete the installation.

Step 2: Open Wireshark and Start the Capture

1. Launch Wireshark.
2. Select the active network interface (Wi-Fi or Ethernet), identified by the moving activity graph.
3. Click the blue shark-fin icon to start capturing. Start the capture before browsing so no traffic is missed.

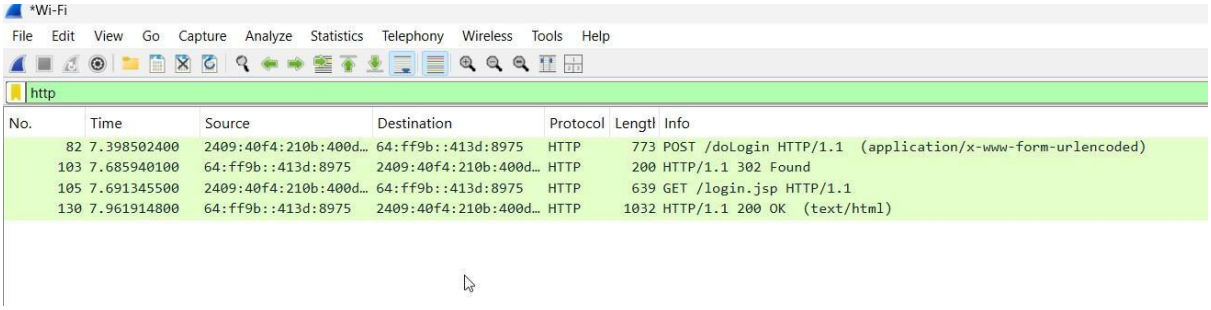
Step 3: Open testfire.net

1. In the browser, visit http://testfire.net.
2. Click **Sign In**.
3. Enter the demo credentials (for example, username: admin, password: admin) and click **Login**.
4. Login was wrong. The wireshark will record the process.



Step 4: Stop the Capture

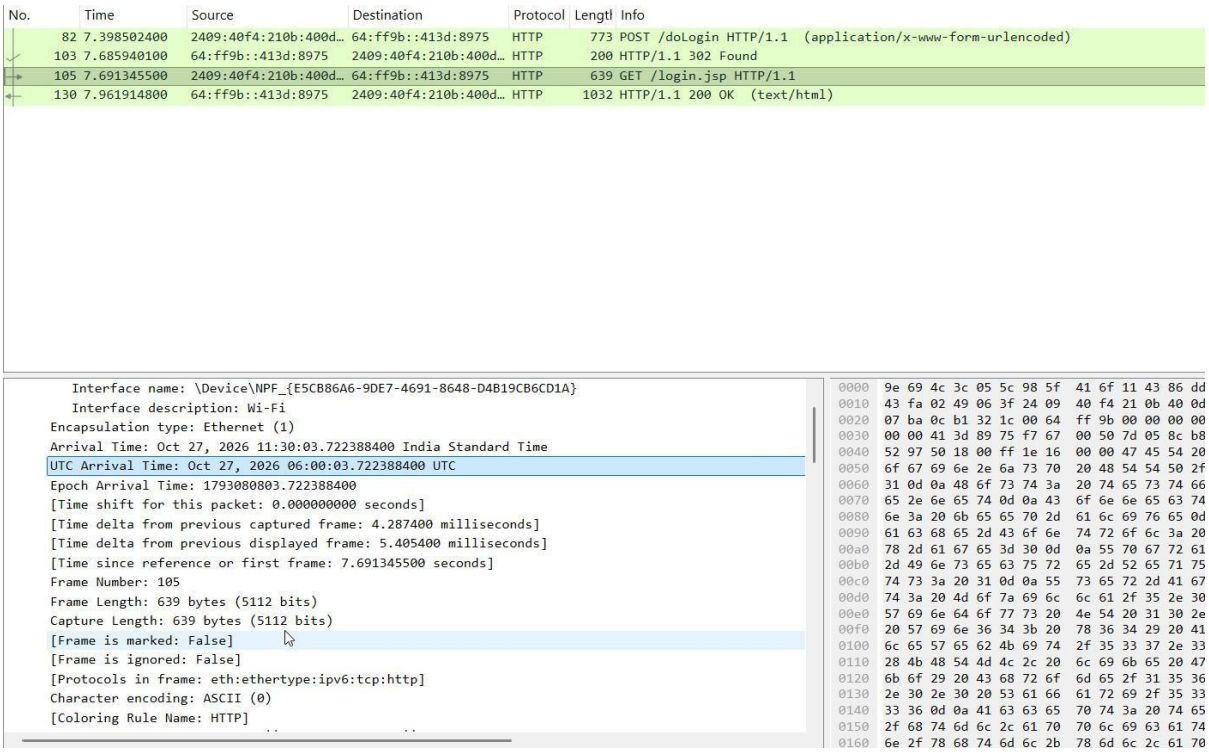
1. Return to Wireshark and click the red Stop button.
2. Save the capture using File > Save As (format: .pcapng).

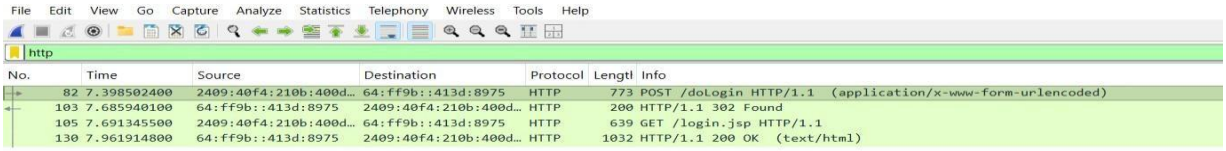


Step 5: Analyze the Captured Packets

After stopping the capture, analyzed the packets in Wireshark

Frame Details. I checked the frame number, the packet length, and the arrival time to know when the packet was captured and how large it was.

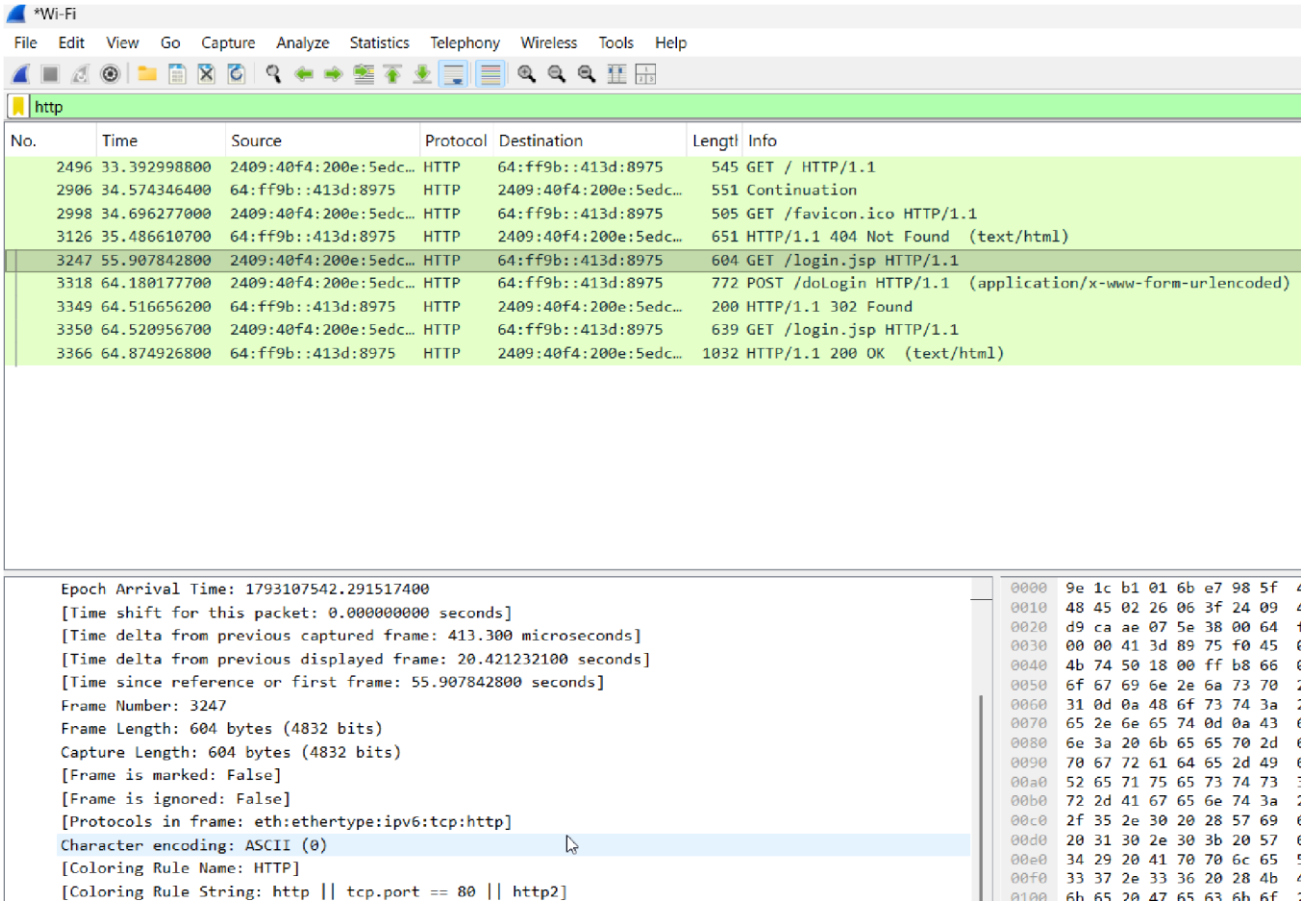




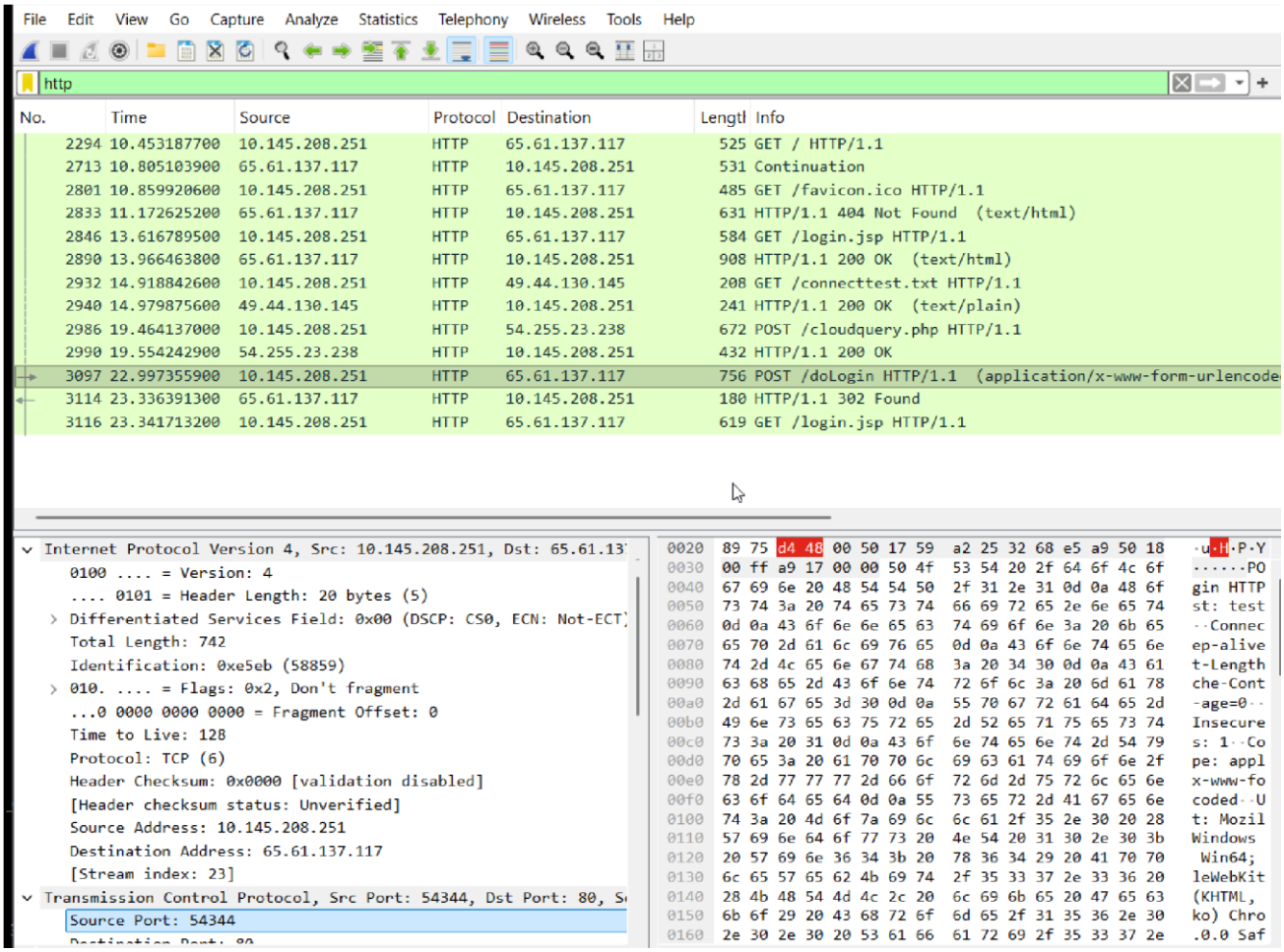


IP Details. I noted the source and destination IP addresses to identify which device sent the packet and which device received it. I also checked the protocol field to confirm whether the packet used TCP, UDP, or ICMP.

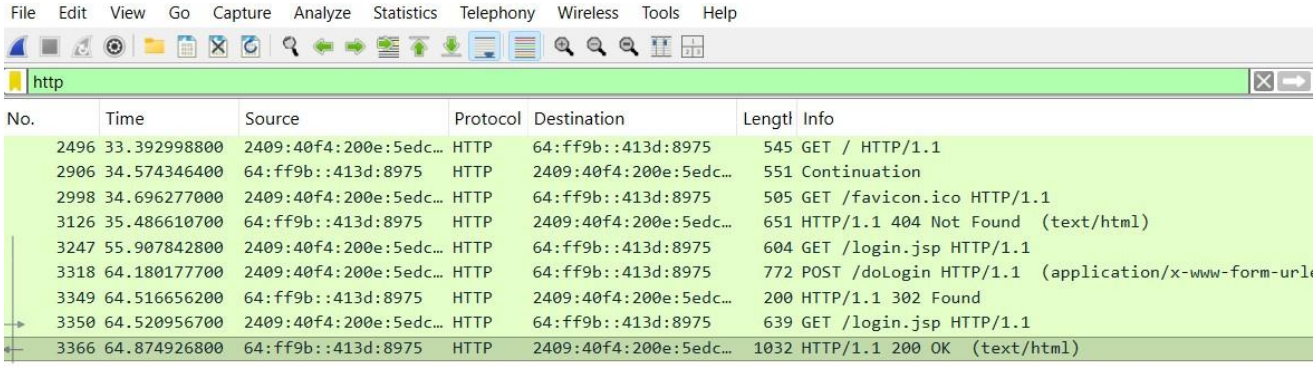
Port Numbers. I checked the source and destination port numbers in the TCP or UDP section. Port 80 indicates HTTP traffic, port 443 indicates HTTPS traffic, and port 53 indicates DNS traffic.

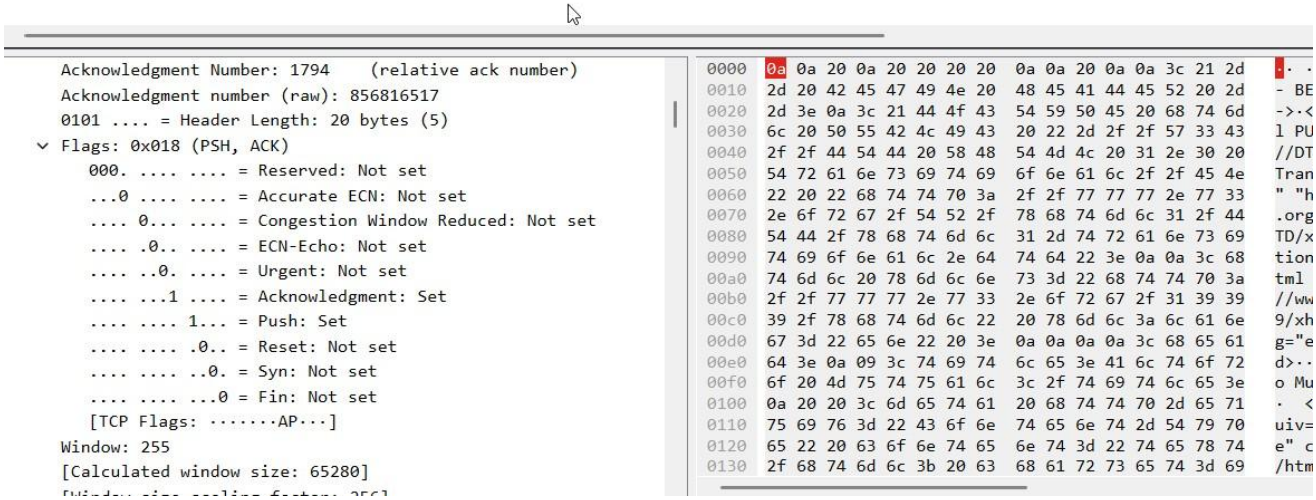


TTL (Time To Live). I checked the TTL value in the IP header, which shows how many routers a packet can still pass through before it is discarded. A TTL of 64 or below usually indicates a Linux system, a value above 64 up to 128 indicates Windows, and a value above 128 indicates a network device such as a router (this is only an estimate).



TCP Flags. I examined the TCP flags to understand the state of the connection. The SYN, SYN-ACK, and ACK flags showed the three-way handshake being completed, which establishes a connection between my computer and the server.





HTTP Status Codes. For the testfire.net traffic, I applied the http filter and checked the status code in the server's response. A status code of 200 means the request was successful, and a code of 404 means the requested page was not found.

